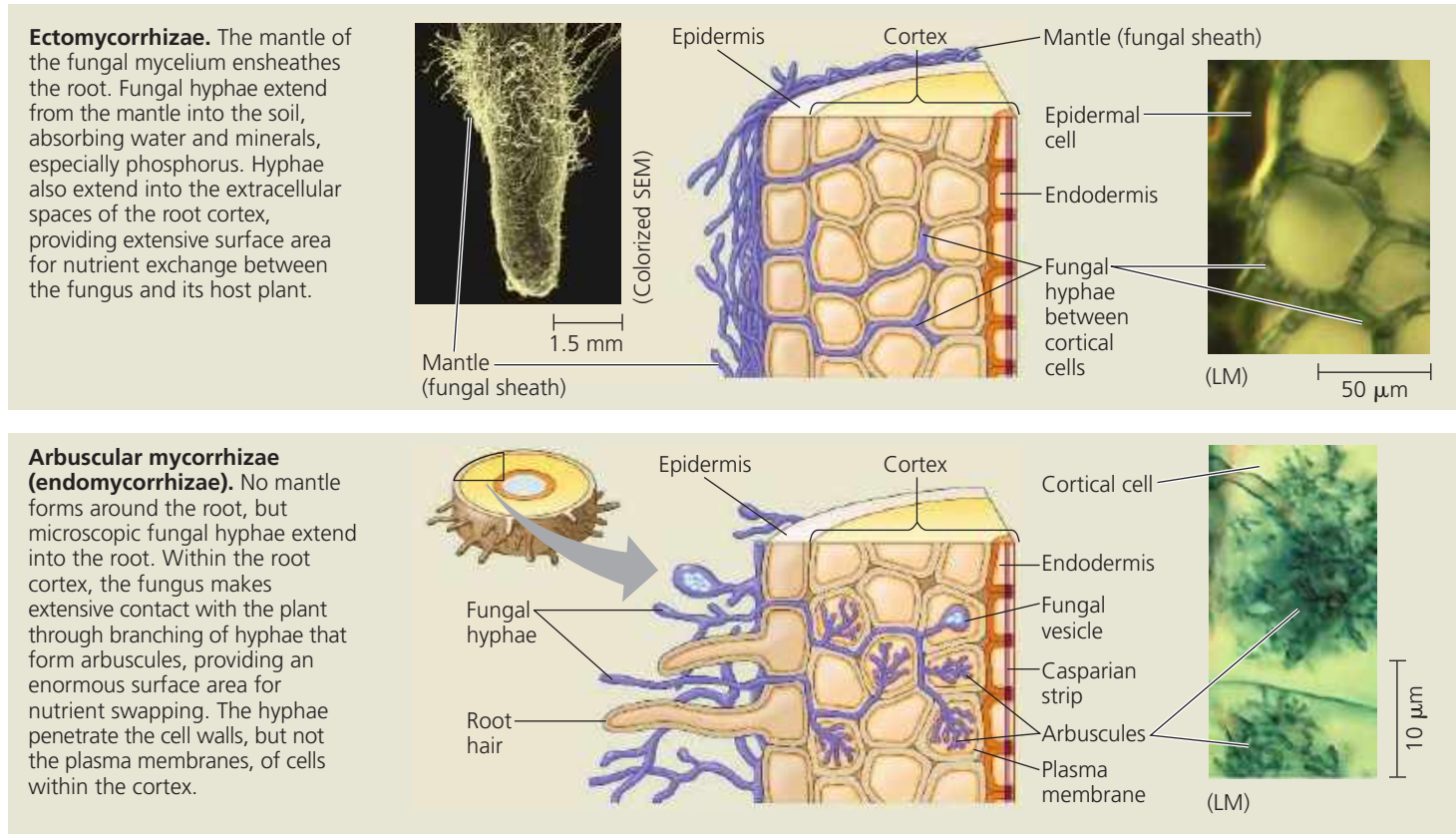


▼ **Figure 37.14 Mycorrhizae.**



but don't pierce the plasma membrane. Instead of entering the cytoplasm, a hypha grows into a tube formed by invagination of the root cell's membrane. This invagination is like poking a finger gently into a balloon without popping it; your finger is like the fungal hypha, and the balloon skin is like the root cell's membrane. After the hyphae have penetrated in this way, some of them branch densely, forming structures called arbuscules ("little trees"), which are important sites of nutrient transfer between the fungus and the plant. Within the hyphae themselves, oval vesicles may form, possibly serving as food storage sites for the fungus. Arbuscular mycorrhizae are far more common than ectomycorrhizae, being found in over 85% of plant species, including most crops. About 5% of plant species don't form mycorrhizal associations. **Figure 37.14** provides an overview of mycorrhizae.

Agricultural and Ecological Importance of Mycorrhizae

Good crop yields often depend on the formation of mycorrhizae. Roots can form mycorrhizal symbioses only if exposed to the appropriate species of fungus. In most ecosystems, these fungi are present in the soil, and seedlings develop mycorrhizae. But if crop seeds are collected in one environment and planted in foreign soil, the plants may show signs of malnutrition (particularly phosphorus deficiency) resulting from

the absence of fungal partners. Treating seeds with spores of mycorrhizal fungi can help seedlings form mycorrhizae, facilitating recovery of damaged natural ecosystems (see Concept 55.5) and improving crop yield.

Mycorrhizal associations are also important in understanding ecological relationships. Arbuscular mycorrhizae fungi exhibit little host specificity; a single fungus may form a shared mycorrhizal network with several plants, even plants of different species. Mycorrhizal networks in a plant community may benefit one plant species more than another. Other examples of how mycorrhizae may affect the structures of plant communities come from studies of introduced plant species. For instance, garlic mustard (*Alliaria petiolata*), a European species that has been introduced into the forests of the eastern half of the United States, does not form mycorrhizae but hinders the growth of other plant species by preventing the growth of arbuscular mycorrhizal fungi.

Epiphytes, Parasitic Plants, and Carnivorous Plants

Almost all plant species have mutualistic relationships with soil fungi, bacteria, or both. Some plant species, including epiphytes, parasites, and carnivores, have unusual adaptations that facilitate exploiting other organisms (**Figure 37.15**).